

Henry Hengyuan Zhao

+65-9135-0337 | hengyuan.z@u.nus.edu | <https://zhaohengyuan1.github.io/>

 Google Scholar |  zhaohengyuan1 |  LinkedIn

Singapore

INTRODUCTION

My research focuses on **multimodal LLMs Post-Training and Interactive Intelligence, AI Agents**. I have several **agent studies** (Chart2Code ACL 2026, WorldGUI ACL 2025 REALM, InterFeedback EMNLP 2025), spanning GUI automation, interactive Feedback learning, and visual chart coding. These studies include designing rigorous evaluation methodologies and building agents with self-verification/critic loops.

Previously, I interned at Sea AI Lab and Microsoft Research, focusing on **post-training techniques** (Genixer ECCV 2024, LOVA3 NeurIPS 2024, etc.), including how to construct the training data, how to synthesize data, and how to design the training framework. I have conducted experiments on both **SFT** (Sea) and **RLVR** (Microsoft) post-training during these two internships. During the internship at **Microsoft**, I trained a multi-role cotrain framework with RLVR and studied test-time scaling by training reward models on Qwen3-VL.

EDUCATION

- **National University of Singapore** Jan 2022 - Present
Ph.D., Electrical Computer Engineering Singapore
- **Nanjing University of Posts and Telecommunications** 2016 - 2020
B.S., Communication Engineering China

EXPERIENCE

- **StepFun Infra Team** April 2026 - present
External Collaborator, AI Agents, Memory System Design w/ Terrance, Alex
Working on designing an industry-level memory management system for the AI Infra.
- **Shanghai AI Lab** Oct 2025 - present
External Collaborator, RLVR Post-Training, Process-Reward Model Design w/ AI4Sci Team
Working on Agentic RL training techniques for serveral post-training projects.
- **Microsoft AI** Aug 2025 - June 2026
Intern, GRPO Post-Training, Reward Model Training w/ Zhengyuan Yang, remote
Working on designing multi-role cotrain framework with GRPO and multimodal reward model training.
- **Sea AI Lab** June 2023 - March 2024
Research Intern, MLLMs, SFT Post-Training, Data Synthesis w/ Pan Zhou, Singapore
Working on multimodal synthetic data generation and MLLMs post-training algorithm development.
- **Alibaba DAMO Academy** Dec. 2021 - March 2023
Research Intern, PEFT for Large Vision Transformers Singapore
Develop PEFT algorithms for Vision Transformers.

PUBLICATIONS

- Wenqi Pei*, **Henry Hengyuan Zhao*** (Equal First Author), Yilai Liu*, Jiahao Meng, CHEN Han, Ziyu Wang, Hongyang Du, "TempCloze: Can Video-LLMs Identify the Missing Middle?", **EMNLP 2026 Findings**.
Summary: In this paper, we systematically design three key time-sensitive video temporal reasoning tasks to investigate existing SOTA LMMs. Our results show a large gap between standard semantic reasoning task with time-sensitive tasks.
- Xiangyu Zhao, **Henry Hengyuan Zhao**, Yiheng Wang, Wanghan Xu, Yuhao Zhou, Qinglong Cao, Zhiwang Zhou, LEI BAI, Wenlong Zhang, Xiao-Ming Wu, "Sci-PRM: A Tool-Aware Process Reward Model for Scientific Reasoning Verification", **KDD 2026**.
Summary: We present a tool-augmented scientific dataset, SCIPRM70K. We propose an automated labeling framework that distinguishes tool selection, calling, and result utilization errors. We train a reward model that scores partial scientific reasoning trajectories and supports test-time selection and self-correction, improving reliability and reducing hallucinations
- Jiahao Tang*, **Henry Hengyuan Zhao*** (Project Lead, Equal First Author), Lijian Wu, Zijian Zhang, Yifei Tao, Dongxing Mao, Wanyang, Jingru Tan, Min Zeng, Min Li, Alex Jinpeng Wang, "From Charts to Code: A Hierarchical Benchmark for Multimodal Models", **ACL 2026**.
Summary: A really challenging benchmark for frontier LMMs, "Chart2Code", which spans three task levels including Chart Reproduction, Chart Editing, and Long-Table to Chart. It includes 2,222 tasks across 22 chart types, and we conducted large-scale experiments on 29 open/closed-source leading models. It provides several insights and findings for chart-to-code research.

- **Henry Hengyuan Zhao**, Kaiming Yang, Wendi Yu, Difei Gao, and Mike Zheng Shou, "WorldGUI: An Interactive Benchmark for Desktop GUI Automation from Any Starting Point", **ACL 2025 REALM**.
Summary: A general agent framework built on a critic/self-verification loop with three key critical designs: Post-Planning, Pre-Action Validation, and Post-Action Critic. We also propose a task state-aware benchmark for testing GUI agents in dynamic, realistic environments.
- **Henry Hengyuan Zhao**, Wenqi Pei, Yifei Tao, Mike Zheng Shou, "InterFeedback: Unveiling Interactive Intelligence of Large Multimodal Models via Human Feedback", **EMNLP 2025 Findings**.
Summary: Study whether frontier multimodal LLMs can improve through real-time human feedback loops. Key findings: (1) accuracy alone may not fully reflect model intelligence; (2) models may over-cater to human preferences rather than genuinely correct errors; (3) low-quality feedback can degrade performance more than simple binary (0/1) signals – directly informing how human-in-the-loop feedback should be designed for guiding/verifying AI systems.
- **Henry Hengyuan Zhao**, Pan Zhou, Difei Gao, Zechen Bai, Mike Zheng Shou, "LOVA³: Learning to Visual Question Answering, Asking and Assessment", **NeurIPS 2024**.
Summary: Only answering questions? Let's think about asking and assessing questions when training MLLMs? Without hyperparameter tuning or additional data annotation, consistent performance improvements are achieved!
- **Henry Hengyuan Zhao**, Pan Zhou, Mike Zheng Shou, "Genixer: Empowering Multimodal Large Language Model as a Powerful Data Generator", **ECCV 2024**.
Summary: How do MLLMs perform in data generation? Proposed the first framework turning MLLMs into scalable data generators, producing 1M samples. We also present two data generators and two MLLMs.
- **Henry Hengyuan Zhao**, Pichao Wang, Yuyang Zhao, Hao Luo, Fan Wang, Mike Zheng Shou, "SCT: A Simple Baseline for Parameter-Efficient Fine-Tuning via Salient Channels", **IJCV 2023**.
Summary: We found that tuning only a small number of task-specific channels, referred to as salient channels, is sufficient. This work represents a remarkable reduction of $780 \times$ in parameter costs compared to its full fine-tuning counterpart.
- **Henry Hengyuan Zhao**, Hao Luo, Yuyang Zhao, Pichao Wang, Fan Wang, Mike Zheng Shou, "Revisit Parameter-Efficient Transfer Learning: A Two-Stage Paradigm", **Arxiv, 2023**. **Efficient Fine-tuning Algorithm**.
- **Hengyuan Zhao**, Xiangtao Kong, Jingwen He, Yu Qiao and Chao Dong, "Efficient Image Super-Resolution using Pixel Attention", **ECCV AIM Challenge 2020**, **600+ citations**, **300+ Github Stars**. **Efficient Model Architecture**.
- Yihao Liu*, **Hengyuan Zhao***, Kelvin CK Chan, Xintao Wang, Chen Change Loy, Yu Qiao and Chao Dong, "Temporally Consistent Video Colorization with Deep Feature Propagation and Self-regularization Learning", **Computational Visual Media, 2024**, **Honorable Mention Award**.
- Yihao Liu, **Hengyuan Zhao**, Jinjin Gu, Yu Qiao, Chao Dong, "Evaluating the Generalization Ability of Super-resolution Networks", **TPAMI 2023**.
- Xiangtao Kong, **Hengyuan Zhao**, Qiao Yu and Chao Dong, "ClassSR: A General Framework to Accelerate Super-Resolution Networks by Data Characteristic", **CVPR 2021**, **300+ Github Stars**.
- **Hengyuan Zhao***, Wenhao Wu*, Yihao Liu*, Dongliang He, "Color2Embed: Fast Exemplar-Based Image Colorization using Color Embeddings", **Arxiv 2021**.
- **Hengyuan Zhao**, Wenze Shao, Bingkun Bao and Haibo Li, "A Simple and Robust Deep Convolutional Approach to Blind Image Denoising", **ICCV Workshop 2019**.

SERVICE CONTRIBUTIONS

Journal Reviewer: International Journal of Computer Vision (IJCV), The IEEE Transactions on Image Processing (TIP).

Conference Reviewer: ICLR 2027, AAAI 2027, NeurIPS 2026, ICML 2026, ICLR 2026, ECCV 2026, ACM MM 2026, NeurIPS 2025, ACM MM 2025, ACL 2025, WACV 2024, 2025, AAAI 2025

Regular reviewer for top-tier venues in machine learning, computer vision, and natural language processing.

TALKS

- I gave an Oral Presentation in the ICLR 2025 Bi-Align Workshop.
- I gave a talk about "Memory Efficient Techniques" at the Show Lab internal meeting.
- I gave a talk on AIM 2020 Efficient Super-Resolution Challenges.

AWARDS

- ICML 2026 Silver Reviewer Awards.
- Honorable Mention Award, Computer Visual Media 2024.